



Southeast University

Introduction to Software Engineering

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Outline

Part 0. Overview

Part 1. Software Processes

Part 2. Modeling

Part 3. Quality Assurance

❖ **Chap 15-16. Software Quality**

❖ **Chap 17-19. Testing Strategy & Techniques**

❖ **Chap 20-21. Security Engineering & SCM**

Part 4. Software Project Management

Chap 15-16. Software Quality

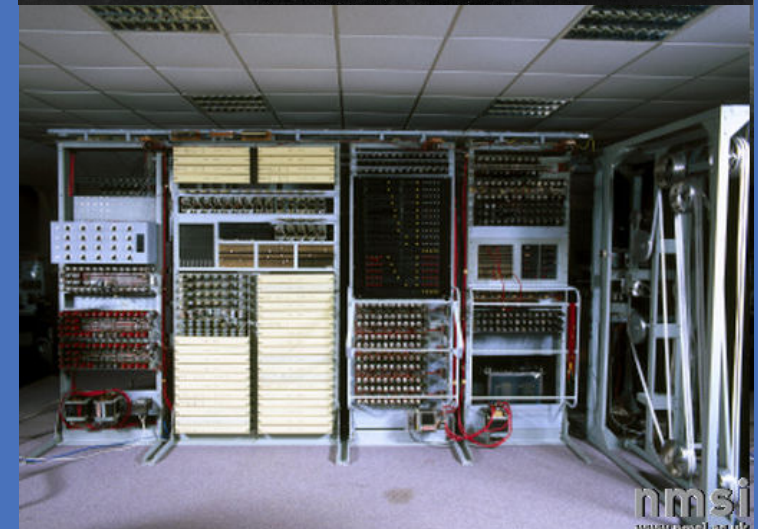
- ❖ 1. The importance of Software Quality
- ❖ 2. What is Quality?
- ❖ 3. Software Quality Standards
- ❖ 4. “Good Enough” Software
- ❖ 5. The Cost of Quality
- ❖ 6. Achieving Software Quality
- ❖ 7. Software Quality assurance



Grace Murray Hopper(1906-1992)
 计算机科学家 美国海军将军
 编译器的发明者
COBOL语言的开发负责人
ACM Grace Murray Hopper Award

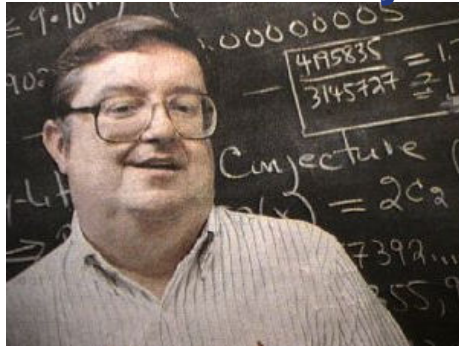
First Computer Bug (1947)

Action started
 stopped - action ✓
 1300 (032) HP-MC 1.982677900 9.03784
 (033) PRO 2 2.130476415 4.61:
 cond 2.130676415
 Relays 6-2 in 033 failed special speed test
 in relay 10.000 test
 Relays changed
 Started Cosine Tape (Sine check)
 Started Multi Adder Test
 Relay #70 Panel
 (moth) in relay.
 First actual case of bug being found.
 action started.
 closed down.



Pentium FDIV bug (1994)

Thomas Nicely



(4195835/3145727) *3145727-4195835

4195835.0/3145727.0 = 1.333 820 449 136 241 002 (Correct value)

4195835.0/3145727.0 = 1.333 739 068 902 037 589 (Returned by flawed Pentium)

4195835*3145727/3145727 = 4195835 (Correct value)

4195835*3145727/3145727 = 4195579 (The flawed Pentium)

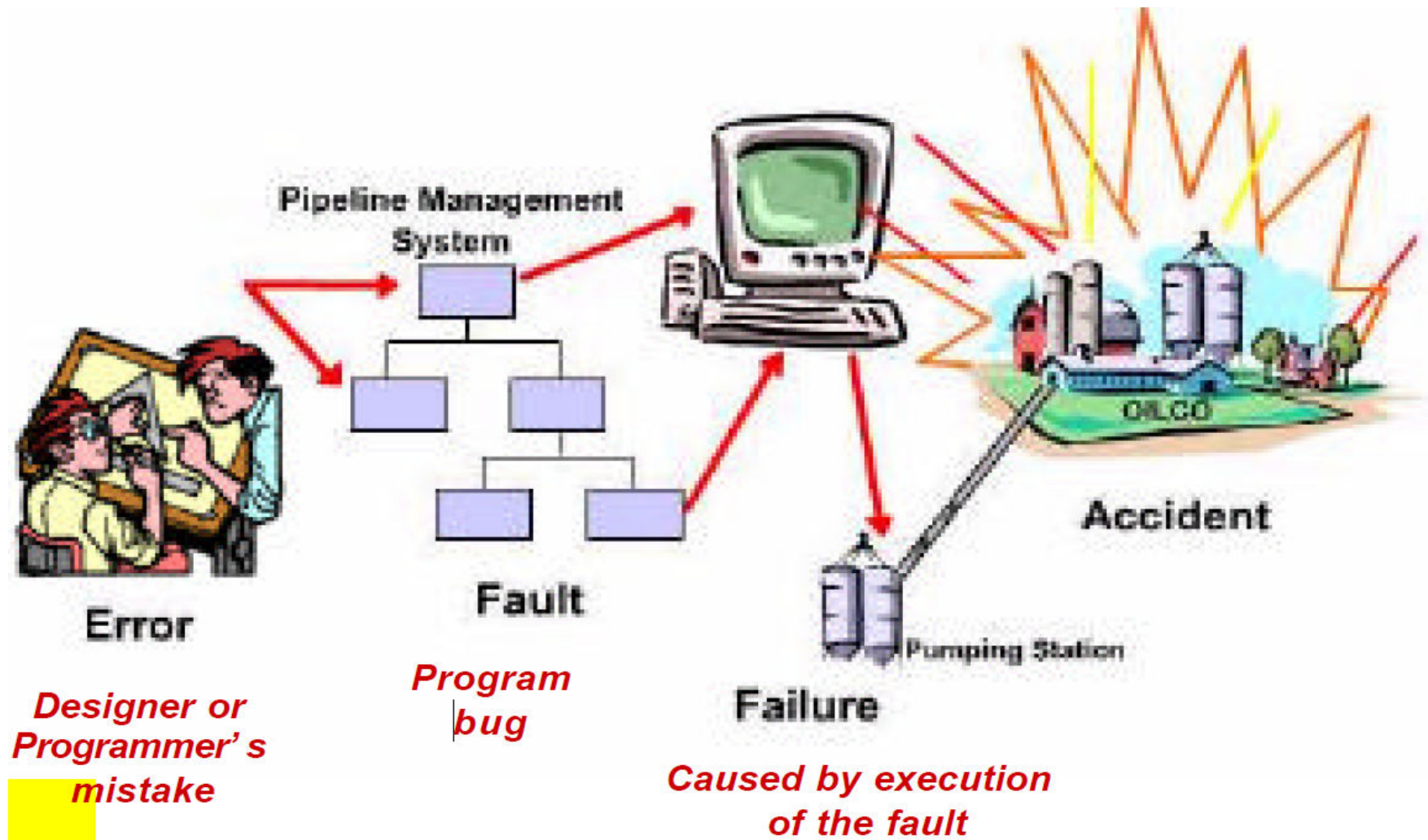
http://en.wikipedia.org/wiki/Pentium_FDIV_bug

Mars Polar Lander (1999)



Nancy G. Leveson, The Role of Software in Spacecraft Accidents

Quality



Quality—A Philosophical View

- ❖ Robert Persig [Per74] commented on the thing we call *quality*:
 - Quality . . . you know what it is, yet you don't know what it is. But that's self-contradictory. But some things are better than others, that is, they have more quality. But when you try to say what the quality is, apart from the things that have it, it all goes poof! There's nothing to talk about. But if you can't say what Quality is, how do you know what it is, or how do you know that it even exists? If no one knows what it is, then for all practical purposes it doesn't exist at all. But for all practical purposes it really does exist. What else are the grades based on? Why else would people pay fortunes for some things and throw others in the trash pile? Obviously some things are better than others . . . but what's the betterness? . . . So round and round you go, spinning mental wheels and nowhere finding anyplace to get traction. What the hell is Quality? What is it?

Quality

❖ The *American Heritage Dictionary* defines *quality* as

▪ “a characteristic or attribute of something.”

❖ For software, two kinds of quality may be encountered:

▪ **Quality of design** encompasses requirements, specifications, and the design of the system.

▪ **Quality of conformance** is an issue focused primarily on implementation.

❖ **User satisfaction = compliant product + good quality + delivery within budget and schedule**

- 《美国传统词典》将质量定义为“某物的特性或属性”。
 - 对于软件，可能会遇到两种质量：
 - 设计质量涵盖需求、规格和系统设计。
 - 符合性质量主要关注实现问题。
- 用户满意度 = 符合产品 + 良好的质量 + 在预算和时间内交付

Software Quality

❖ Software quality can be defined as:

- *An effective software process applied in a manner that creates a useful product that provides measurable value for those who produce it and those who use it.*

1. 什么是软件质量——一个有效的软件过程，其应用方式创造了一个有用的产品，为生产它的人和使用它的人提供可衡量的价值。

McCall's Triangle of Quality

2. 产品转移

1. 可移植性
2. 可复用性
3. 互操作性

1. 产品修改

1. 维护性
2. 灵活性
3. 易测试性

Maintainability

Flexibility

Testability

Portability

Reusability

Interoperability

PRODUCT REVISION

PRODUCT TRANSITION

PRODUCT OPERATION

Correctness

Usability

Efficiency

Reliability

Integrity

3. 产品运行

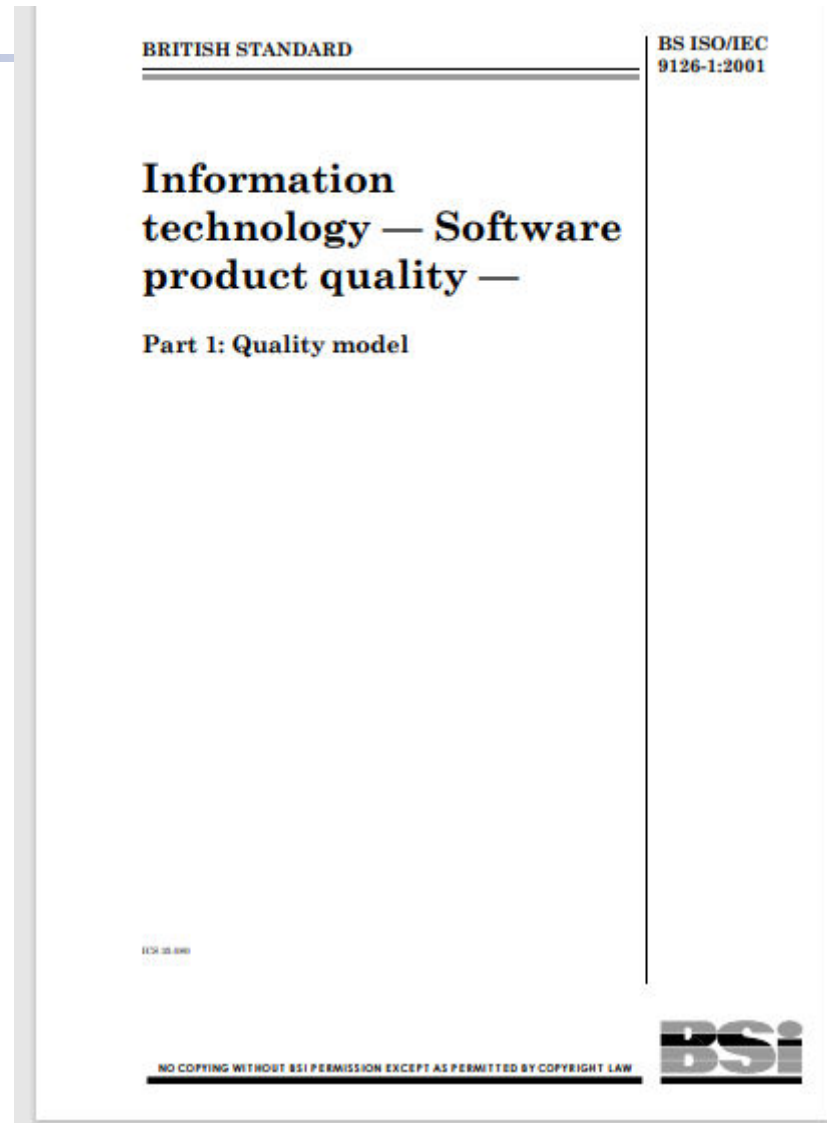
1. 正确性
2. 易用性
3. 效率
4. 可靠性
5. 完整性

ISO 9126

- ❖ **Functionality**
- ❖ **Reliability**
- ❖ **Usability**
- ❖ **Efficiency**
- ❖ **Maintainability**
- ❖ **Portability**

4. ISO9126质量因素

- 功能性
- 可靠性
- 易用性
- 效率
- 维护性
- 可移植性



ICS 35.080
L 77



中华人民共和国

GB/T 25000.1—2010

软件工程 软件产品质量
(SQuaRE) SQuaRE

Software engineering—Software product Quality
Evaluation (SQuaRE)—Guide to SQuaRE

(ISO/IEC 25000:2005, IDT)

2010-09-02 发布

中华人民共和国国家质量监督检验检疫总局
中国国家标准化管理委员会

ICS 35.080
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中华人民共和国

GB/T 16260.2—2006/ISO

软件工程 产品质量
第2部分：外部度量

Software engineering—
Product quality—
Part 2: External metrics

(ISO/IEC TR 9126-2:2003, IDT)

2006-03-14 发布

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中华人民共和国国家标准

GB/T 16260.3—2006/ISO/IEC TR 9126-3:2003

软件工程 产品质量
第3部分：内部度量

Software engineering—
Product quality—
Part 3: Internal metrics

(ISO/IEC TR 9126-3:2003, IDT)

2006-03-14 发布

2006-07-01 实施

中华人民共和国国家质量监督检验检疫总局 发布
中国国家标准化管理委员会

Measuring Quality

- ❖ General quality dimensions and factors are not adequate for assessing the quality of an application in concrete terms.
- ❖ Project teams need to develop **a set of targeted questions to assess** the degree to which each application quality factor has been satisfied.
- ❖ Subjective measures of software quality may be viewed as little more than personal opinion.
- ❖ **Software metrics** represent indirect measures of some manifestation of quality and attempt to quantify the assessment of software quality.

-
- 一般的质量维度和因素不足以具体评估应用程序的质量。
 - 软件质量的主观测量可能只被视为个人意见。
 - 项目团队需要制定一套有针对性的问题，以评估每个应用质量因素的满足程度。
 - 软件度量代表某些质量表现的间接测量，并尝试量化软件质量的评估。

The Software Quality Dilemma (困境)

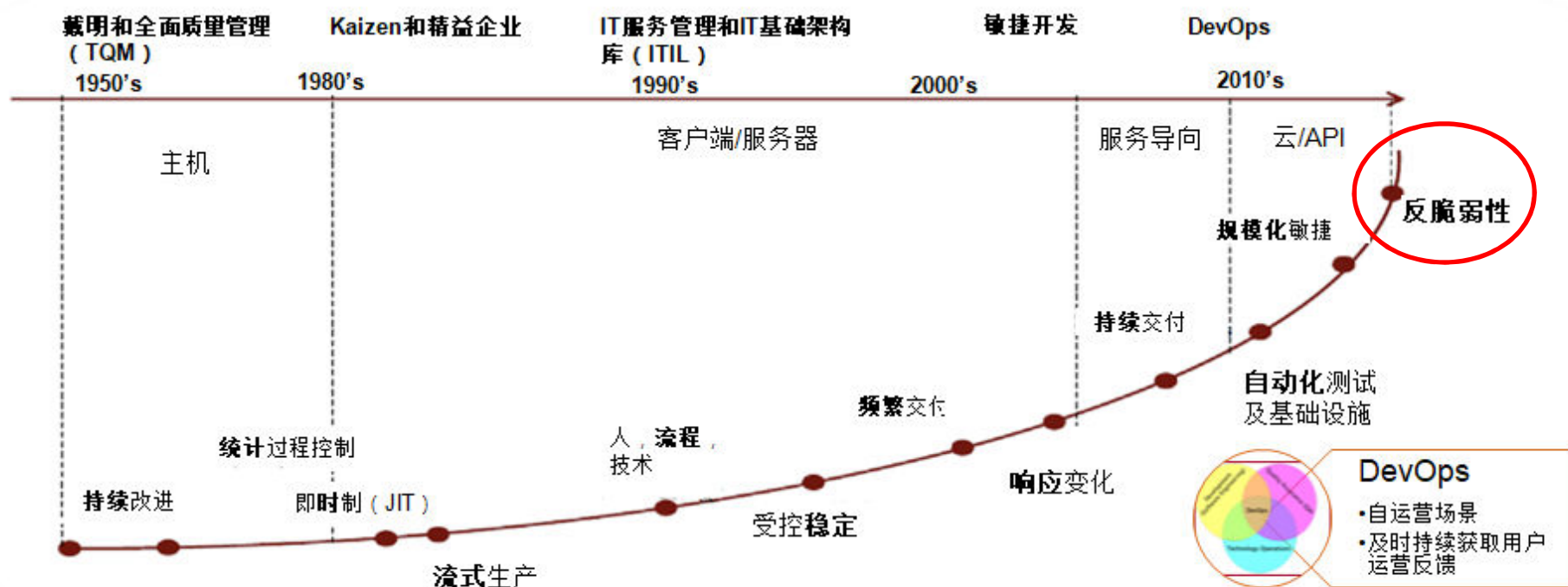
- ❖ If you produce a software system that has terrible quality, you lose because no one will want to buy it.
- ❖ If on the other hand you spend infinite time, extremely large effort, and huge sums of money to build the absolutely perfect piece of software, then it's going to take so long to complete and it will be so expensive to produce that you'll be out of business anyway.
- ❖ Either you missed the market window, or you simply exhausted all your resources.
- ❖ So people in industry try to get to that magical middle ground where the product is **good enough** not to be rejected right away, such as during evaluation, but also not the object of so much perfectionism and so much work that it would take too long or cost too much to complete. [Ven03]

“Good Enough” Software

- ❖ **Good enough software delivers high quality functions and features that end-users desire, but at the same time it delivers other more obscure or specialized functions and features that contain known bugs.**

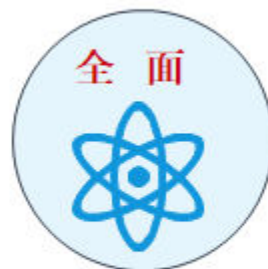
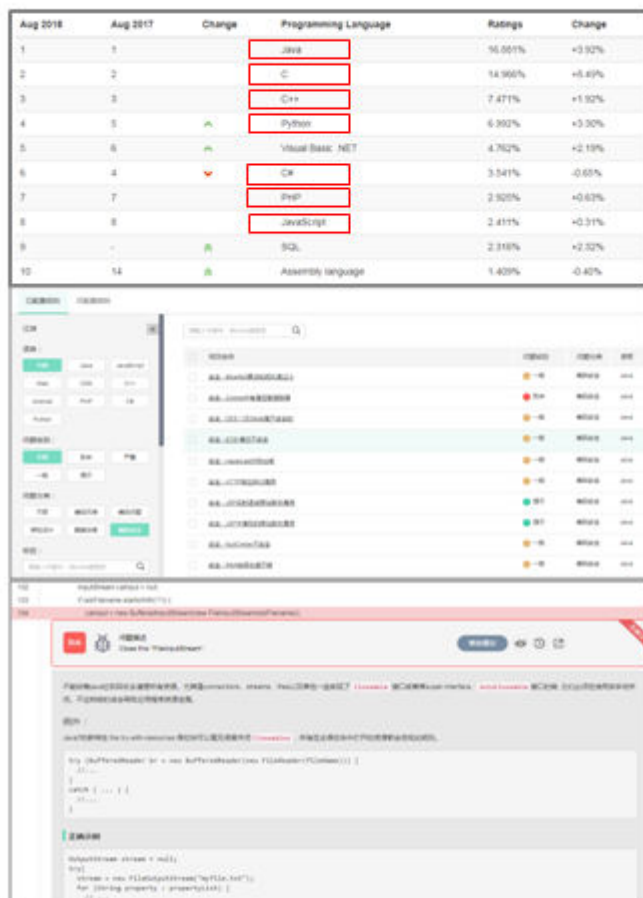
6. “足够好”的软件——解决用户的关键问题、后续修复已知的bug

新形势新挑战促进软件研发模式不断优化创新



- (1) 价值驱动，聚焦最高价值最高优先级的工作； (2) 持续研发，小步快跑，快速闭环；
(3) 拥抱变化，根据市场需要和研发能力按需发布； (4) 客户深度参与，联合创新，运营驱动开发。

代码检查 - 持续看护代码质量, 防止代码腐化



全面

- 覆盖了Top10主流编程语言的其中7种；
- 从7个维度全面评价代码质量：编码安全、编码缺陷、编码风格、架构、依赖包安全、圈复杂度，代码重复。



专业

- 华为30年经验积累，提供专业级的预置规则集；
- 提供专业的缺陷修复建议，指导代码质量改进。



精准

- 提供自定义规则集，可定制企业级质量标准，精准击中企业关心问题；
- 缺陷精确定位到代码行。

“Good Enough” Software

❖ Arguments *against* “good enough.”

- If you work for a small company be wary of this philosophy. If you deliver a “good enough” (buggy) product, you risk **permanent damage to your company’s reputation**.
- You may **never get a chance to deliver version 2.0** because bad buzz may cause your sales to plummet and your company to fold.
- If you work in certain application domains (e.g., real time embedded software, application software that is integrated with hardware can be negligent and **open your company to expensive litigation**（诉讼）.

反对“足够好”的观点:

- 如果你在小公司工作，需警惕这种理念。如果你交付了一个“足够好”的（有缺陷的）产品，你可能永久性地损害公司的声誉。
- 你可能永远没有机会交付2.0版本，因为负面评价可能导致你的销售额骤降，最终导致公司倒闭。
- 如果你在某些特定应用领域工作（例如，实时嵌入式软件，集成硬件的应用软件），一旦出现疏漏，可能会导致公司面临昂贵的诉讼。

Cost of Quality

❖ Prevention Costs (COP)

- training
- standard, procedures
- planning
- quality improvement
- audits
- analysis

❖ Appraisal Costs (COA)

- reviews
- walkthroughs
- testing
- supplier monitoring

7. 质量成本（逐渐增加）

1. 预防成本（COP）
2. 评估成本（COA）
3. 内部失败成本
4. 外部失败成本

Cost of Quality

❖ Internal Failure Costs

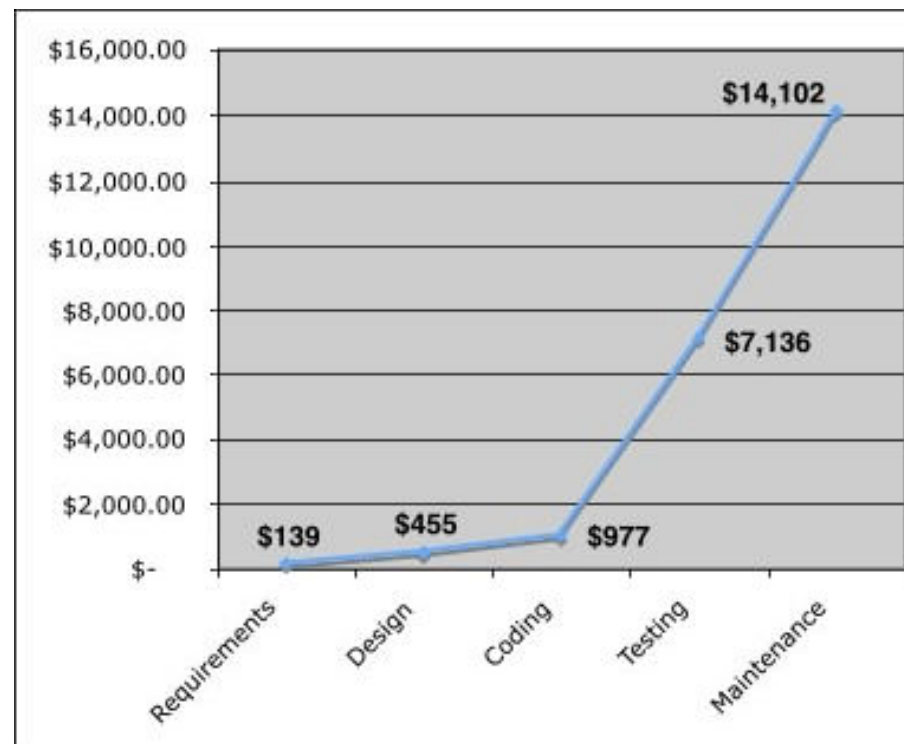
- Correction and re-work
- Regression testing
- Wasted tester time
- Cost of late shipment
- Opportunity cost of late shipment

❖ External Failure Costs

- Technical support calls
- Investigate customer complaints
- Refunds and recalls
- Lost sales
- Lost goodwill
- Penalties
- Liability costs
- Legal fees

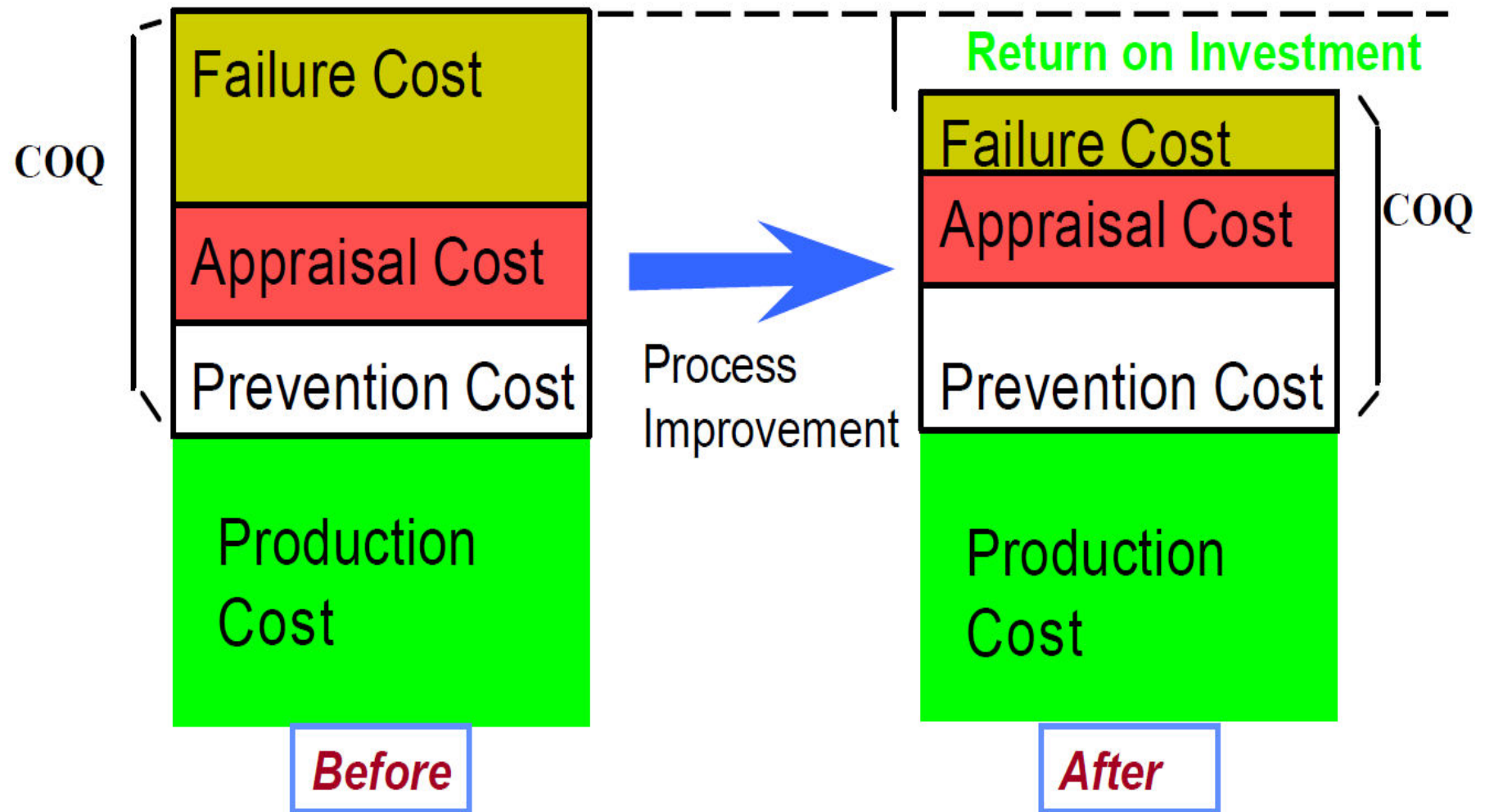
Cost of Quality

- ❖ The relative costs to find and repair an error or defect increase dramatically as we go from prevention to detection to internal failure to external failure costs.

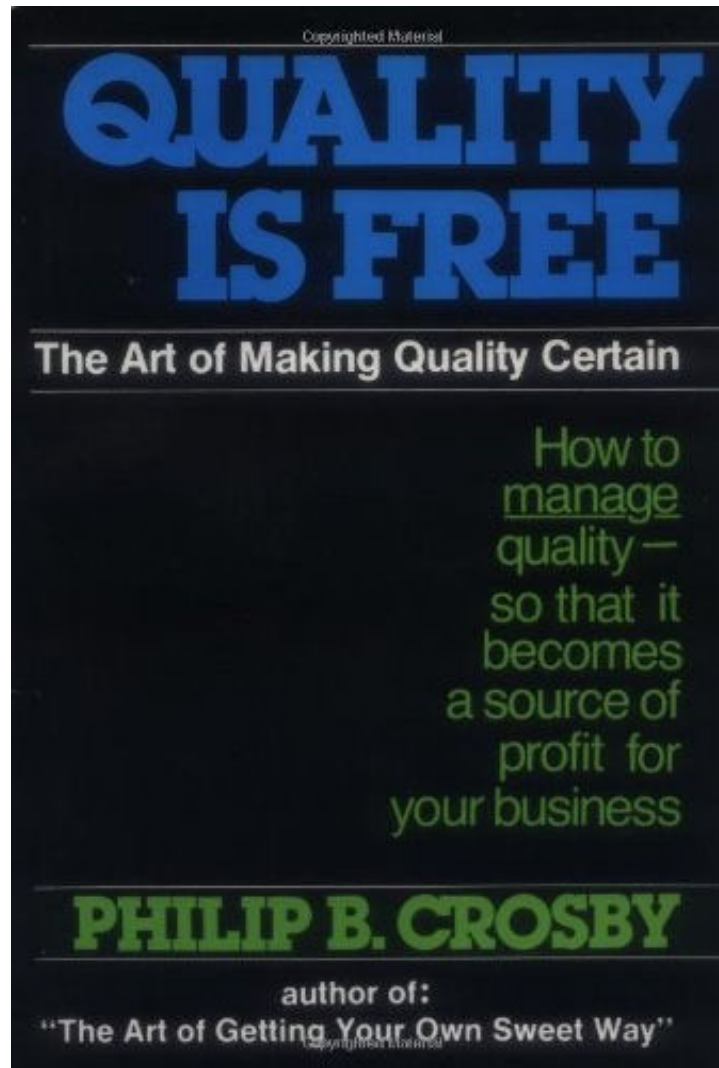


查找和修复错误或缺陷的相对成本在我们从预防到检测到内部故障到外部故障成本时显著增加。

Quality is Free!! 质量是免费的!!



Quality is Free!!



Achieving Software Quality

- ❖ **Software quality is the result of good project management and solid engineering practice.**
- ❖ **To build high quality software you must understand the problem to be solved and be capable of creating a quality design that conforms to the problem requirements.**
- ❖ **Eliminating architectural flaws during design can improve quality.**

- 软件质量是良好的项目管理和扎实的工程实践的结果。
- 要构建高质量的软件，你必须了解要解决的问题，并能够创建符合问题要求的质量设计。
- 在设计过程中消除架构缺陷可以提高质量。

Achieving Software Quality

❖ **Project management** – project plan includes explicit techniques for quality and change management.

❖ **Quality control** - series of inspections, reviews, and tests used to ensure conformance of a work product to its specifications.

❖ **Quality assurance** - consists of the auditing and reporting procedures used to provide management with data needed to make proactive decisions.

- 项目管理——项目计划包括明确的质量和变更管理技术。
- 质量控制——一系列检查、审查和测试，用于确保工作产品符合其规格。
- 质量保证——由审计和报告程序组成，用于向管理层提供做出积极决策所需的数据。

Elements of SQA

❖ **Standards**

❖ **Reviews and Audits**

❖ **Testing**

❖ **Error/defect collection and analysis**

❖ **Change management**

❖ **Education**

❖ **Vendor management**

❖ **Security management**

❖ **Safety**

❖ **Risk management**

1. 标准

2. 审查和审核

3. 测试

4. 错误/缺陷收集和分析

5. 变更管理

6. 教育

7. 供应商管理

8. 安全管理

9. 安全性

10. 风险管理

3. SQA小组的作用

Role of the SQA Group

❖ Prepares an SQA plan for a project.

- The plan identifies
 - **evaluations** to be performed
 - **audits and reviews** to be performed
 - **standards** that are applicable to the project
 - **procedures** for error reporting and tracking
 - **documents** to be produced by the SQA group
 - amount of **feedback** provided to the software project team

❖ Participates in the development of the project's software process description.

- The SQA group **reviews the process description** for compliance with organizational policy, internal software standards, externally imposed standards (e.g., ISO-9001), and other parts of the software project plan.

1. 为项目准备 SQA 计划

2. 审查**过程描述**是否符合组织政策、内部软件标准、外部强制标准（例如，ISO-9001）和软件项目计划的其他部分

Role of the SQA Group

- ❖ **Reviews software engineering activities to verify compliance with the defined software process.**
- ❖ **Audits designated software work products to verify compliance with those defined as part of the software process.**
- ❖ **Ensures that deviations in software work and work products are **documented** and handled according to a documented procedure.**
- ❖ **Records any noncompliance and **reports** to senior management.**

3. 审查**软件工程活动**是否符合定义的软件过程
4. 审核指定的**软件工作产品**是否符合定义为软件过程的一部分的产品
5. 确保软件工作和工作产品中的偏差被记录并根据文件化程序进行处理。
6. 记录任何违规行为并向高级管理层报告

SQA Goals

- ❖ Requirements quality.
- ❖ Design quality.
- ❖ Code quality.
- ❖ Quality control effectiveness.

4. SQA的目标

1. 需求质量
2. 设计质量
3. 代码质量
4. 质量控制的有效性

Statistical SQA

**Product
& Process**

**Collect information on all defects
Find the causes of the defects
Move to provide fixes for the process**

measurement

统计SQA (软件质量保证)

- 收集所有缺陷的信息
- 查找缺陷的原因
- 提供过程修复的措施
-理解如何提高质量.....

*... an understanding of how
to improve quality ...*



Southeast University



Thank You !

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